

Ashish Ranjan

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OBJECTIVE

Final-year CSE (AI & ML) student at VIT Bhopal University with a strong foundation in C++, DSA, Machine Learning and Java full-stack web development. Experienced in building end-to-end applications using Java, Spring Boot, React, and MySQL. Seeking opportunities in Software Engineer and Machine Learning Engineer roles to contribute to scalable enterprise solutions while growing within a structured organization.

EDUCATION

VIT Bhopal University <i>Bachelor of Technology – Computer Science & Engineering (AI & ML)</i> • CGPA: 8.86 / 10.0	Sehore, MP 2023 – 2027
Dr. Jagannath Mishra College <i>Intermediate (Class XII)</i> • Percentage: 78.2%	Muzaffarpur, Bihar 2021 – 2022
Disha Delphi Public School <i>Matriculation (Class X)</i> • Percentage: 83.6%	Kota, Rajasthan 2019 – 2020

TECHNICAL SKILLS

Backend Frameworks: Spring Boot, Spring MVC, REST APIs, Flask
Frontend: React.js, JavaScript, HTML, CSS
Languages: Java, Python, C/C++, JavaScript, SQL
Databases: MySQL (Relational), JDBC, JPA/Hibernate (Basic)
Libraries & Tools: NumPy, Pandas, Scikit-learn, Matplotlib, Tableau
DevOps & Version Control: Git, GitHub, VS Code, IntelliJ IDEA
DSA: Arrays, Linked Lists, Trees, Sorting & Searching
Relevant Coursework: Data Structures & Algorithms, DBMS, OS, OOP, Cloud Computing, Software Engineering
Soft Skills: Adaptability, Communication, Teamwork, Attention to Detail

EXPERIENCE

MPOnline Limited – Bhopal, MP <i>Software Development & AI/ML Trainee</i> • Developed 3+ full-stack applications by integrating responsive front-end interfaces with relational databases and modular backend logic • Applied SQL and PL/SQL concepts including CRUD operations, joins, stored procedures, functions, and triggers for efficient database management • Built end-to-end machine learning workflows using 5+ Python libraries (NumPy, Pandas, Matplotlib, Scikit-learn, and Pickle), covering preprocessing, feature engineering, model training, evaluation, and inference • Currently working on model serialization using Pickle/Joblib to deploy trained machine learning models as web applications using Flask • Following industry-standard development practices including Git-based version control, debugging, code documentation, testing, and collaborative project management	May 2026 – Present
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PROJECTS

Hospital Management System <i>Java, Spring Boot, Spring Security, JWT, MySQL, React, REST API</i> • Built a full-stack Hospital Management System with secure JWT-based authentication and role-based authorization for Admin, Doctor, Receptionist, and Patient using Spring Security. • Created RESTful APIs for patient management, doctor management, appointment scheduling, billing, and medical records using Spring Boot, Spring Data JPA, Hibernate, and MySQL. • Implemented end-to-end appointment workflows including booking, rescheduling, cancellation, doctor availability, and appointment status tracking to streamline hospital operations.

- Designed modules for patient medical history, diagnoses, prescriptions, lab reports, invoices, and payment history with optimized relational database design.
- Structured the application using a layered architecture (Controller, Service, Repository) with input validation, centralized exception handling, Swagger API documentation, and Git-based version control.

Student Library Management System | *Java, Spring Boot, MySQL, REST API, React*

- Built a full-stack **CRUD** application using **Java Spring Boot** as the backend REST API layer and React.js for the frontend interface.
- Created RESTful endpoints (GET, POST, PUT, DELETE) for managing book inventory, student records, and borrowing history using **Spring MVC** and **JDBC/MySQL**.
- Applied input validation, exception handling, and HTTP status codes following industry REST API standards.
- Structured the project using **MVC architecture** with separation of Controller, Service, and Repository layers.
- *Tech Stack: Java, Spring Boot, MySQL, React.js, REST API, Git*

Sentiment Analyzer | *Java, Spring Boot, HuggingFace API, HttpClient, Python, MySQL, JUnit 5*

- Integrated **HuggingFace Inference API** using Java’s HttpClient to classify text as Positive, Negative, or Neutral with confidence scores.
- Built a **Spring Boot REST endpoint** that accepts text input, calls the ML API, parses JSON responses, and returns structured sentiment results.
- Processed a dataset of **500+ product reviews** (Yelp CSV) using OpenCSV, analyzed them in batches, and stored results in MySQL.
- Executed **batch processing logic** with rate-limiting and retry mechanisms to handle API timeouts and quota limits gracefully.
- Wrote **unit tests** using JUnit 5 for sentiment parsing and text preprocessing logic.

Real-Time Collaboration Platform | *JavaScript, HTML, CSS, Python, Flask*

- Designed and developed a responsive web platform for hackathon team coordination with real-time task management.
- Implemented filtering algorithms to filter results based on user requirements.
- Used Flask as a lightweight backend to serve APIs for task state management.
- *Tech Stack: JavaScript, HTML, CSS, Python, Flask*

CERTIFICATIONS

Machine Learning with Python – University of Michigan — Certificate Link	
<i>Skills: Scikit-Learn, Regression, Classification, Clustering, Model Evaluation</i>	<i>Completed: Dec 2025</i>
NPTEL – Cloud Computing (Elite Grade)	
<i>Cloud architectures, deployment models, scalable system design</i>	<i>2025</i>
NPTEL – Internet Of Things (overall score 90)	
<i>IoT architecture , embedded systems , IoT communication protocols</i>	<i>2026</i>
Deloitte Australia – Data Analytics Job Simulation — Certificate Link	
<i>Tableau Dashboard, Excel data classification, business insights</i>	<i>July 2025</i>
Deloitte Australia – Technology Job Simulation — Certificate Link	
<i>Development and coding simulation</i>	<i>July 2025</i>

AREAS OF INTEREST

Java Full Stack Development		Cloud Engineering		Machine Learning		Data Structures and Algorithms
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